

Fast Fourier Transform Algorithms - DIFFFT

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Decimation In Frequency FFT (DIFFFT) Algorithm

- Decompose the computation into successively smaller DFT computations.
- We exploit both the symmetry and periodicity property of the complex exponential $W_N^{nk} = e^{-j\left(\frac{2\pi}{N}\right)nk}$
- In this algorithm the decomposition is based on the output sequence $X(k)$ into successively smaller subsequences

Let N be equal to 2^v , where v is an integer

Consider a finite length sequence $x(n)$

Let $L \leq N$ be the length of the signal $x(n)$

Then we can define N -pt DFT of $x(n)$ as

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{nk}, \quad k=0, 1, \dots, N-1 \quad \text{--- (1)}$$

Even numbered frequency points can be written as

$$X(2r) = \sum_{n=0}^{N-1} x(n) W_N^{n2r}, \quad k=0, 1, \dots, \frac{N}{2}-1 \quad \text{--- (2)}$$

Odd frequency points can be written as

$$X(2r+1) = \sum_{n=0}^{N-1} x(n) W_N^{(2r+1)n}, \quad k=0, 1, 2, \dots, \frac{N}{2}-1 \quad \text{--- (3)}$$

consider eqn (2), $X(2r) = \sum_{n=0}^{N-1} x(n) W_N^{2rn}$

We can expand it as follows

$$X(2r) = \sum_{n=0}^{\frac{N}{2}-1} x(n) W_N^{2nr} + \sum_{n=\frac{N}{2}}^{N-1} x(n) W_N^{2nr} \quad \text{--- (4)}$$

$$= \sum_{n=0}^{\frac{N}{2}-1} x(n) W_N^{2nr} + \sum_{n=0}^{\frac{N}{2}-1} x(n+\frac{N}{2}) W_N^{2r(n+\frac{N}{2})} \quad \text{--- (5)}$$

$$W_N^{2r(n+\frac{N}{2})} = W_N^{(2rn+rN)} = W_N^{2rn} \cdot W_N^{rN} = W_N^{2rn} = W_{N/2}^{rn} \quad \text{--- (6)}$$

$\therefore$ Eqn (5) can be written as

$$X(2r) = \sum_{n=0}^{\frac{N}{2}-1} x(n) \cdot W_{N/2}^{rn} + \sum_{n=0}^{\frac{N}{2}-1} x(n+\frac{N}{2}) W_{N/2}^{rn}$$

$$X(2r) = \sum_{n=0}^{\frac{N}{2}-1} \underbrace{[x(n) + x(n+\frac{N}{2})]}_{g(n)} W_{N/2}^{rn} \quad \text{--- (7)}$$

Consider now the odd numbered frequency points

$$X(2r+1) = \sum_{n=0}^{N-1} x(n) W_N^{(2r+1)n} \quad r=0, 1, \dots, N/2-1$$

$$X(2r+1) = \sum_{n=0}^{N/2-1} x(n) W_N^{(2r+1)n} + \sum_{n=N/2}^{N-1} x(n) W_N^{(2r+1)n} \quad \text{--- (8)}$$

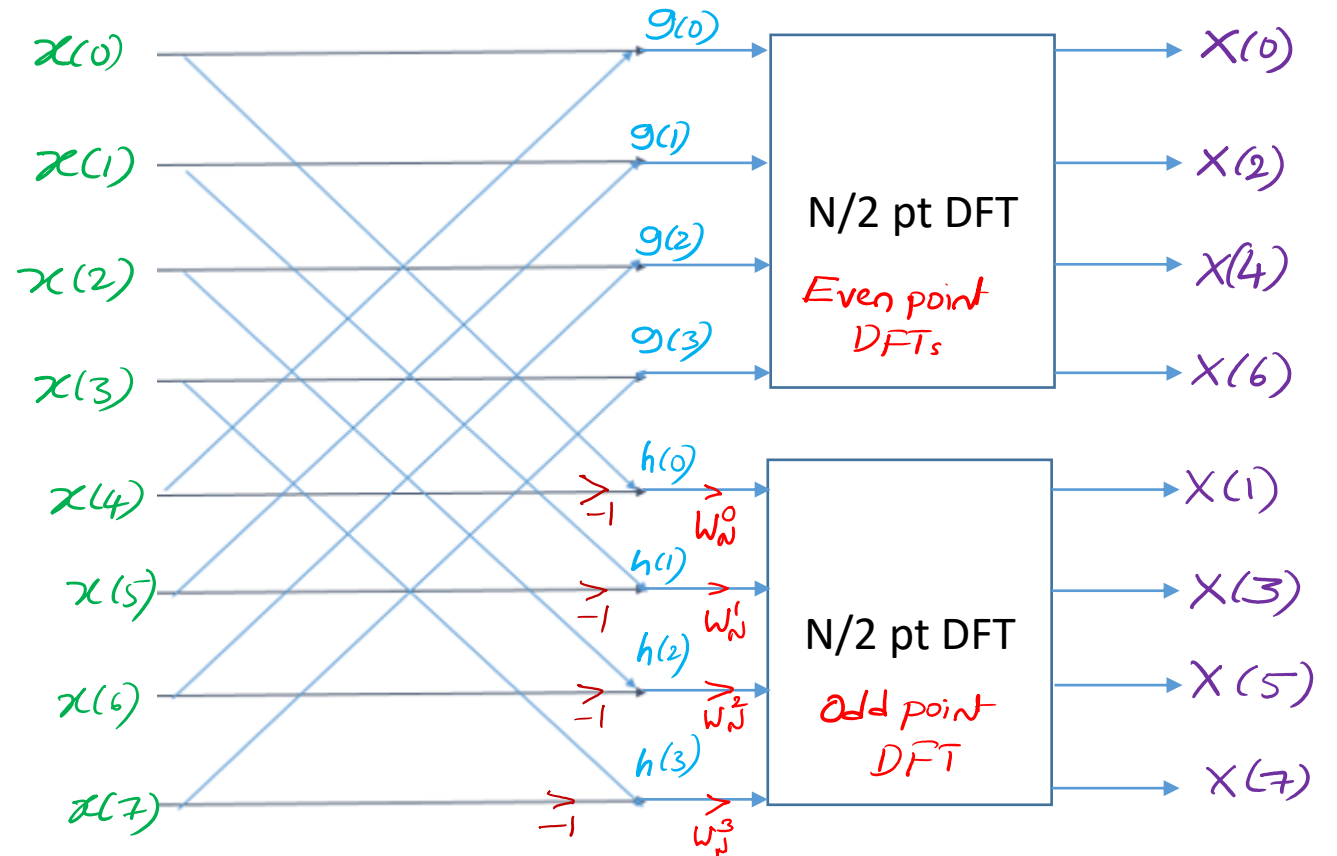
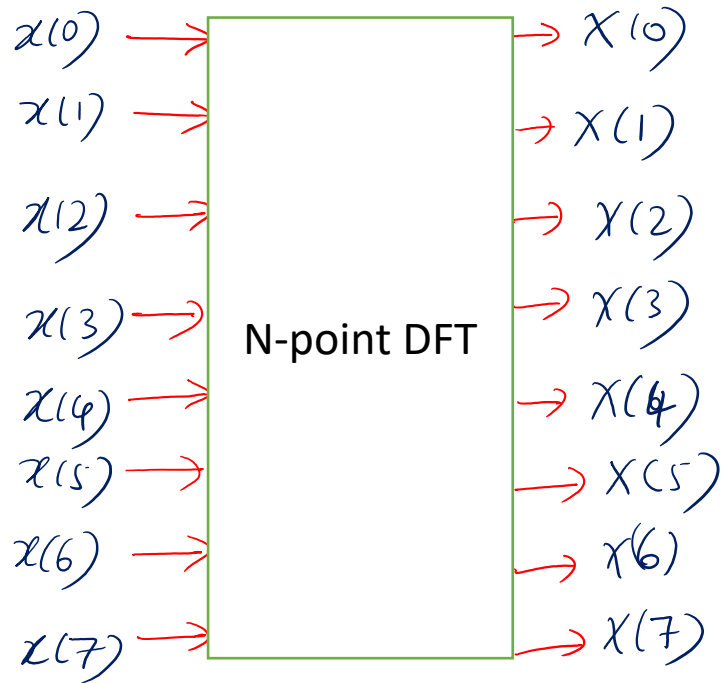
Consider the second term in (8)

$$\begin{aligned} \sum_{n=N/2}^{N-1} x(n) W_N^{(2r+1)n} &= \sum_{n=0}^{N/2-1} x(n + \frac{N}{2}) W_N^{(2r+1)(n + \frac{N}{2})} \\ &= \sum_{n=0}^{N/2-1} x(n + \frac{N}{2}) W_N^{2rn} \cdot W_N^{2r \frac{N}{2}} \cdot W_N^n \cdot W_N^{N/2} \\ &= W_N^{2r \frac{N}{2}} \cdot W_N^{N/2} \cdot \sum_{n=0}^{N/2-1} x(n + \frac{N}{2}) W_N^n \cdot W_{N/2}^{rn} \\ &= \underbrace{(1)} \cdot \underbrace{(-1)} \sum_{n=0}^{N/2-1} x(n + \frac{N}{2}) W_N^n \cdot W_{N/2}^{rn} \quad \text{--- (9)} \end{aligned}$$

Therefore eqn (8) can be rewritten as

$$X(2r+1) = \sum_{n=0}^{N/2-1} x(n) W_N^{(2r+1)n} + (-1) \sum_{n=0}^{N/2-1} x(n+N/2) W_N^{(2r+1)n} \quad \text{--- (10)}$$

$$X(2r+1) = \sum_{n=0}^{N/2-1} \underbrace{[x(n) - x(n+N/2)]}_{h(n)} W_N^n \cdot W_{N/2}^r \quad \text{--- (11)}$$



Equations (7) and equation (11) can be rewritten as

$$G(r) = X(2r) = \sum_{n=0}^{N/2-1} g(n) W_{N/2}^{rn} \quad \text{--- (12)}$$

$$\& X(2r+1) = \sum_{n=0}^{N/2-1} h(n) W_N^n \cdot W_{N/2}^{rn} \quad \text{--- (13)}$$

Now we consider eqn (12)

$$G(l) = \sum_{n=0}^{N/2-1} g(n) W_{N/2}^{ln}$$

Then even DFT points can be

$$G(2r) = \sum_{n=0}^{N/2-1} g(n) W_{N/2}^{2rn} \quad r = 0, 1, \dots, \frac{N}{4}-1 \quad \text{--- (14)}$$

Odd DFT points are

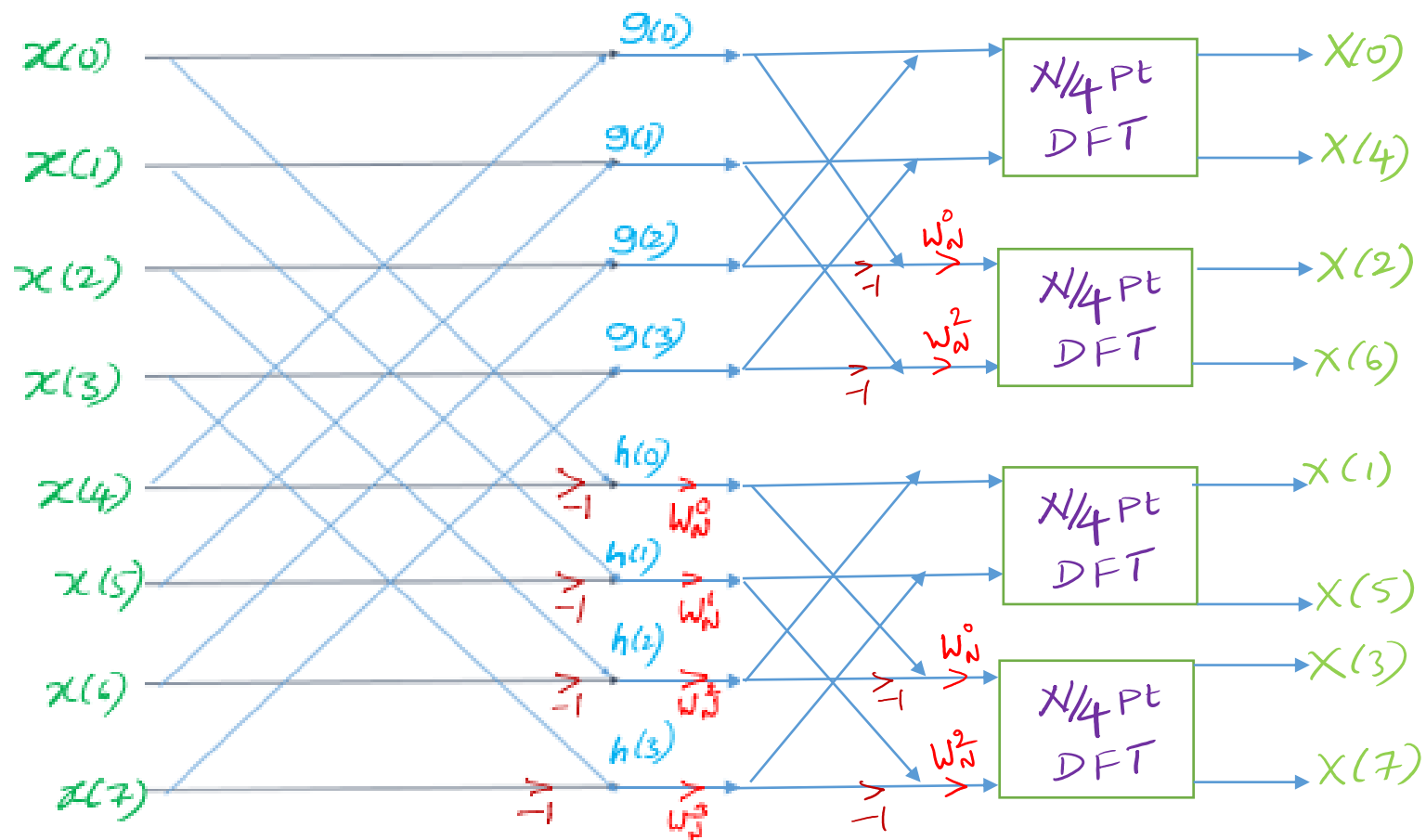
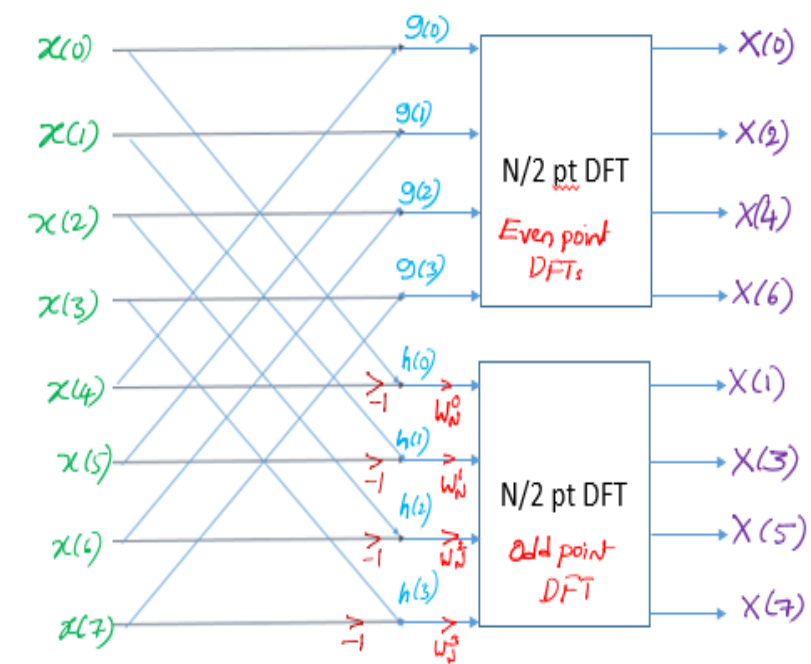
$$G(2r+1) = \sum_{n=0}^{N/2-1} g(n) W_{N/2}^{(2r+1)n} \quad r = 0, 1, \dots, \frac{N}{4}-1 \quad \text{--- (15)}$$

Consider now eqⁿ (14)

$$\begin{aligned}
 G_1(2r) &= \sum_{n=0}^{N/2-1} g(n) W_{N/2}^{2rn} = \sum_{n=0}^{N/4-1} g(n) W_{N/2}^{2rn} + \sum_{n=N/4}^{N/2-1} g(n) W_{N/2}^{2rn} \\
 &= \sum_{n=0}^{N/4-1} g(n) W_{N/4}^{rn} + \sum_{n=0}^{N/4-1} g(n+N/4) W_{N/2}^{2r(n+N/4)} \\
 &= \sum_{n=0}^{N/4-1} g(n) W_{N/4}^{rn} + \sum_{n=0}^{N/4-1} g(n+N/4) W_{N/4}^{rn} \cdot \underbrace{W_{N/2}^{2rN/4}}_1
 \end{aligned}$$

$$G_1(2r) = \sum_{n=0}^{N/4-1} \left[g(n) + g\left(n + \frac{N}{4}\right) \right] W_{N/4}^{rn}$$

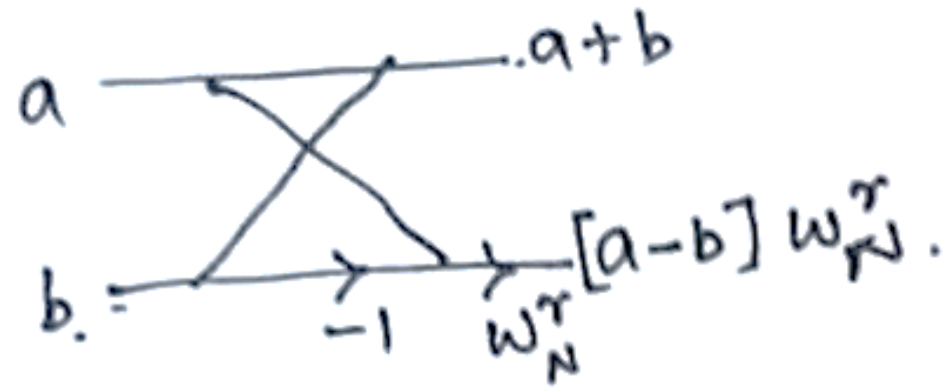
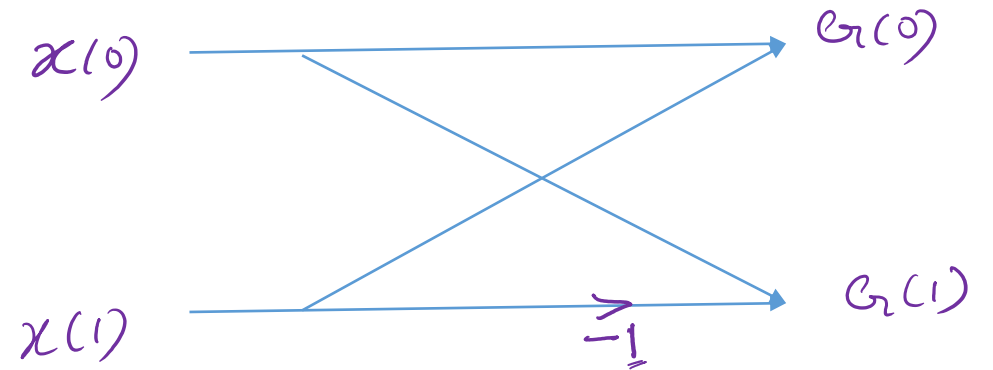
Similarly $G_1(2r+1) = \sum_{n=0}^{(N/4-1)} \left[g(n) - g\left(n - \frac{N}{4}\right) \right] W_{N/2}^n \cdot W_{N/4}^{rn}$

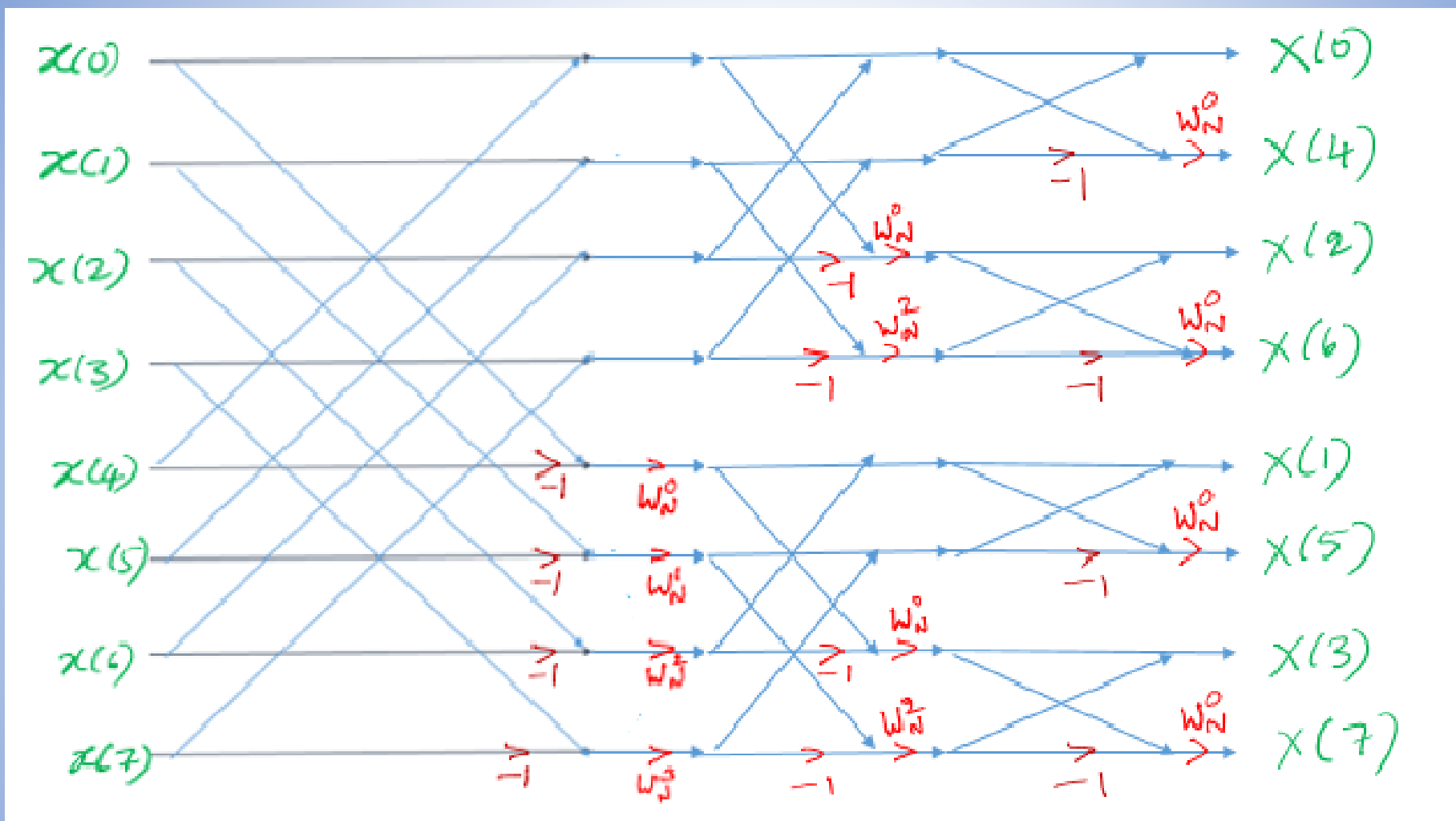


$$G(r) = \sum_{n=0}^{N/4-1} \underbrace{[g(n) + g(n + \frac{N}{4})]}_{x(n)} w_{N/4}^{rn} \rightarrow r = 0, 1, \dots, N/4 - 1$$

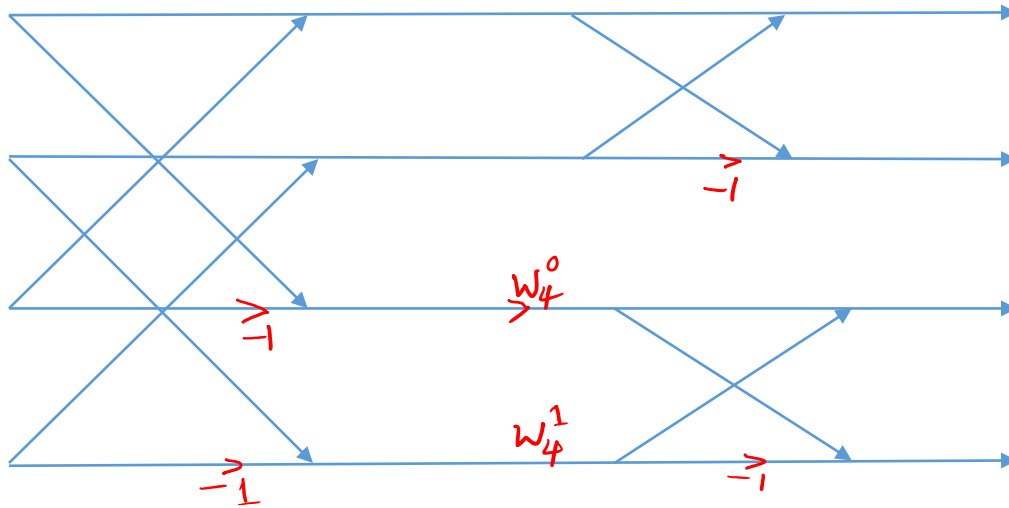
For $N = 8$, $G(r) = \sum_{n=0}^1 x(n) w_2^{rn} \quad r = 0, 1$

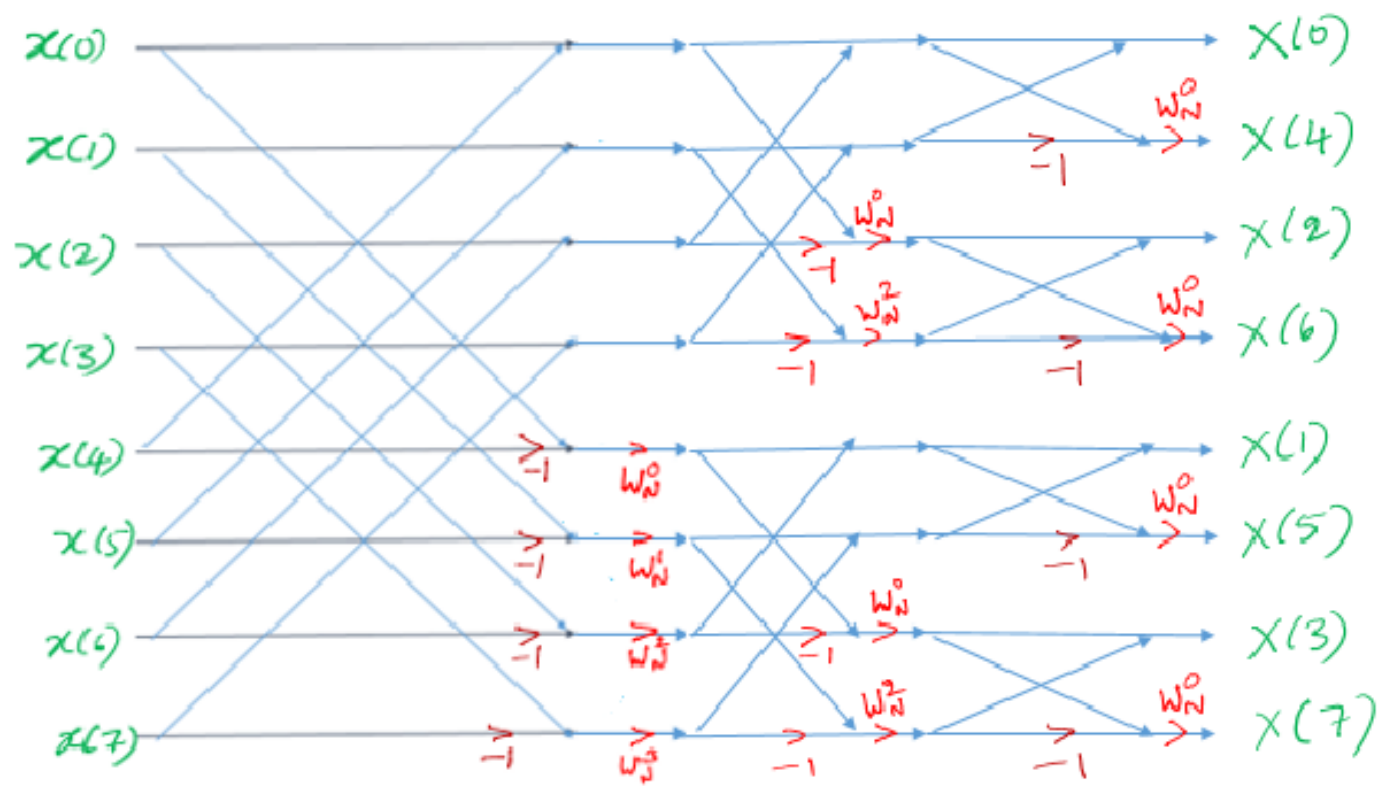
$x(0)$ → $\boxed{\begin{matrix} N/4 \text{ PE} \\ \text{DFT} \end{matrix}}$ → $G(0)$ $G(r) = x(0) w_2^0 + x(1) w_2^r$
 $x(1)$ → $G(1)$ $G(0) = x(0) + x(1)$
 $G(1) = x(0) + x(1) w_2^1$
 $= x(0) - x(1)$





Compute 4-pt DFT of a sequence $x(n) = \{1, 2, 3, 4\}$ using DIF FFT algorithm





Number of stages in the signal flow graph = $\log_2 N$

Number of Butterflies in the signal flow graph = $\frac{N}{2} \log_2 N$

Number of complex additions in each butterfly = 2

Number of complex multiplications in each butterfly = 1

Total number of complex additions and multiplications = ? $(N \log_2 N + \frac{N}{2} \log_2 N)$

*Thank
you*

